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GenapSys: Business Models for the Genome

In April 2013, Hesaam Esfandyarpour, founder and CEO of GenapSys, believed the company was close to launching its first product: a deoxyribonucleic acid (DNA) sequencer known as GENIUS (**G**ene **E**lectronic **N**ano **I**ntegrated **U**ltra-**S**ensitive). Esfandyarpour believed this device would become an industry-leading technology with its low cost and rapid sequencing ability. While sequencers had existed for decades, most were prohibitively expensive for nearly all but the best-funded research institutions, costing hundreds of thousands of dollars on average. GenapSys's device would be less expensive and could sequence a genome within eight hours for around $100.[a]

Esfandyarpour headed to his office in Redwood City, California, for an important meeting with Senior Director of Operations and Strategy Leila Rastegar (HBS '11) where the two would discuss the complex series of tradeoffs they had to consider in selecting which business model to pursue. The company had recently completed its first round of venture capital financing and its investors wanted to review GenapSys's business strategy.

To this end, Esfandyarpour and Rastegar analyzed three options. In the first model, GenapSys would sell sequencers and test cartridges (containing the microchip that DNA was placed on) to the existing market of customers but differentiate on its faster sequencing time and the flexibility this gave customers to pursue additional projects. In the second model, GenapSys would grow the market by selling its machine at a low price, charge more for the test cartridges, and target customers not yet served by the major sequencing companies. The third model had GenapSys sell sequencers and test cartridges at or below cost but collect the data clients produced to create a database of genomic data. GenapSys would make money by charging access fees to researchers and pharmaceutical companies, as well as hosting an online store for genetic tests developed using this information.

Each model had unique opportunities and risks in terms of market opportunity, competition, and company development. Selecting the right model was critical, since Esfandyarpour believed that GENIUS could have an immediate impact on medical research and patient care. "This space is complex and moving quickly. We won't get two chances," Rastegar explained.

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[a] The figures quoted in this case were illustrative and did not necessarily reflect the prices GenapSys would ultimately attach to its commercial device or services.

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From DNA Sequencing to Personalized Medicine

DNA molecules were a key ingredient in the genes that made the human body function, and any abnormalities could produce defective proteins and potentially cause disease.[1] An individual's entire collection of DNA was known as a genome.[2] DNA was composed, in part, of nucleotides, which contained four nitrogen bases: adenine (A), thymine (T), guanine (G), and cytosine (C).[3] These nitrogen bases bonded with one another to form base pairs.[4] Specific arrangements of base pairs comprised genes, which in turn told the body how to make proteins such as hemoglobin and hormones such as estrogen and insulin.[5] Each person had some 20,000 genes, made up of 3 billion total base pairs.[6]

Sequencing was the process of identifying the order of base pairs.[7] Methods varied, but in general, DNA was split into small segments prior to sequencing and multiple copies of each segment were made. Segments were then placed in the sequencer and chemicals were added to identify base pairs. After sequencing, the segments were reassembled and analyzed, often by bioinformatics software.[8]

DNA was sequenced for a variety of purposes. For researchers, DNA provided information about the genetics of various diseases and conditions. Diagnostics companies sequenced DNA when conducting genetic tests to determine whether a patient had a certain condition. Pharmaceutical companies sequenced DNA as part of their research and development (R&D). Sequencing was also important in developing targeted drug therapies by helping researchers understand genetic deficiencies at the molecular level. For example, cancer drugs were being developed to focus on just the molecules involved in tumor development.[9]

In the early 2010s, most sequencing was done for research purposes, accounting for 90% of the total use of sequencers.[10] By 2013, the genomics of some 5,000 genetic diseases were known.[11] From that information, molecular diagnostics companies developed tests to detect some diseases. However, researchers had yet to fully understand the genomics behind diseases involving multiple genes and those which were impacted by lifestyle and environmental influences.[12]

Esfandyarpour noted that most sequencers used optical systems: DNA segments were labeled with dyes, and cameras visually examined it. One observer described the optical sequencing method, which required "flooding DNA snippets with chemicals called reagents that contain one of DNA's four bases. Every base has only one match; when a base finds a match, a light flashes. The sequencer sees it and records the base."[13] The complex nature of optical systems made such devices expensive, as well as labor- and time-intensive to operate.

In addition to the cost of purchasing a sequencer, customers had to buy the reagents necessary to prepare samples. For example, Illumina's HiSeq 2000, a high throughput sequencer,[b] sold for $690,000 and consumed on average $400,000 in reagents annually.[14] Illumina's MiSeq sequencer sold for $125,000 and required $125,000 in reagents each year.[15] Rastegar further noted that customers of these devices had to pay high "instrument services" fees to keep sequencers running at peak performance; the service contract for Illumina's HiSeq machines could cost upwards of $62,000 per year.[16] Spending on consumables (materials used during sequencing) was often higher than spending on machines: in 2011, companies earned $758 million from selling sequencers and $813 million from consumables.[17]

[b] Throughput referred to the number of total base pairs that a machine could read in a single run.

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Personalized Medicine

Sequencing played an important role in personalized medicine.[c] By knowing each patient's genetics, physicians hoped to personalize healthcare to each individual.[18] Identifying diseases and disease risk through genetic testing was a key principle of personalized medicine. By 2013, such tests were available for some 2,000 diseases.[19] "The clinical utility of sequencing DNA will become increasingly apparent as new diagnostic tests and drugs that rely on molecular information come into use; this will compel more clinical institutions to install sequencers," analysts noted.[20] (See **Exhibit 1** for revenues from DNA tests.)

Personalized medicine also had the potential to improve drug development. Even the most effective drugs did not work for many patients: For example, just 25% of patients responded to the bestselling cancer drug at the start of the twenty-first century and just 57% of patients responded to the bestselling diabetes medication.[21] Little had changed a decade later, as analysts noted in 2011 that "seven of the 14 major drug classes are effective in no more than 50% of patients."[22]

Analysts believed that pharmaceutical firms would increasingly develop drugs for specific patient subsets instead of looking for "blockbuster" drugs.[23] Some such drugs were already available, as were tests known as "companion diagnostics" to determine if a patient had the specific mutation that a drug targeted. In 2011, Xalkori and its companion diagnostic were approved to treat patients suffering from late-stage non-small cell lung cancers (NSCLC) with a mutation on the anaplastic lymphoma kinase gene.[24] Just 1% to 7% of patients with NSCLC had this mutation.[25] That same year, Zelboraf and its companion diagnostic were approved to treat late-stage or unresectable melanoma (a skin cancer) for patients with a mutation on the BRAF V600E gene.[26]

Some companies directly targeted consumers. For $99, 23andMe mailed a kit to customers' homes for a saliva sample which the customer sent to the company.[27] From this sample, 23andMe gave customers basic genetic information, such as how they might respond to drugs and what conditions or diseases the customer could pass along to their children.[28] However, 23andMe only accounted for a very small portion (some 0.1%) of a customer's genome.[d]

To make personalized medicine accessible, many parties were working towards sequencing a genome for $1,000. One analyst expected someone to hit this threshold in 2015.[29] While sequencing costs had plummeted in the twenty-first century (see **Exhibit 2**), machine costs remained high, as did costs for genetic tests (ranging up to $8,000). Moreover, it could take weeks for some machines to sequence an entire genome.

Evolution of DNA Sequencing Technology

The first sequencing technology was created in the late 1970s, and named after its creator Frederick Sanger.[30] The Sanger technology and sequencing process was improved upon in successive

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[c] According to the National Human Genome Research Institute, "Personalized medicine is an emerging practice of medicine that uses an individual's genetic profile to guide decisions made in regard to the prevention, diagnosis, and treatment of disease. Knowledge of a patient's genetic profile can help doctors select the proper medication or therapy and administer it using the proper dose or regimen." (Source: "Personalized Medicine," *National Human Genome Research Institute*, www.genome.gov/glossary/index.cfm?id=150, accessed September 2013.)

[d] It was important to note though that the vast majority of human DNA (some 99%) was not involved in the production of protein and was known as noncoding DNA. (Source: Jenny Marder, "Mysterious Noncoding DNA: 'Junk' or Genetic Power Player?" *PBS NewsHour*, November 7, 2011, http://www.pbs.org/newshour/rundown/junk-dna/, accessed May 2015.)

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years through better equipment and methods for preparing and analyzing DNA.[31] However, costs remained high. In 2000, it would have cost approximately $100 million to sequence a complete human genome.[32] "Sanger sequencing instruments have inherent technical limitations that restrict the speed, capacity, and cost-effectiveness of the technology," one analyst noted.[33]

The next wave of machines, known as next-generation sequencing or NGS (see **Exhibit 3** for some examples), emerged in 2005 after the Human Genome Project[e] (HGP) was completed.[34] The HGP used Sanger machines, and the time to sequence the reference human genome took some 13 years and cost $2.7 billion (in 1991 dollars).[35] NGS machines differed from Sanger technology in that they could process more DNA simultaneously and faster than earlier machines.[36] Companies emerging during this time included 454 Life Sciences, Agencourt Personal Genomics, and Solexa, which were acquired by established firms such as Applied Biosystems Inc. (which later became Life Technologies) and Illumina.[37] The valuations associated with those transactions demonstrated the market's potential. For example, Life Technologies acquired Ion Torrent Systems Inc. for $725 million in 2010 just three years after its founding.[38]

In 2013, 5,951 NGS machines were in use.[39] The market was led by Life Technologies and Illumina.[40] (See **Exhibit 4** for the dollar market share of the NGS market by company.) NGS machines could be segmented into larger and more expensive high-throughput machines and smaller, less expensive machines known as benchtop sequencers.[41] Benchtop sequencers generated less data from a single run than higher-throughput devices, but comprised 4,350 of the 5,951 NGS machines in the field.[42] Illumina's MiSeq benchtop sequencer cost $125,000.[43] For smaller-sized customers and those with specific sequencing needs, and for whom price was a key consideration, benchtop sequencers were attractive options.[44]

Analysts grouped the new types of sequencers under development in the early 2010s in a broad category called "emerging," (see **Exhibit 5**).[45] Those sequencers were designed to generate results faster, simplify DNA preparation steps, improve accuracy, and reduce cost.[46] Some emerging devices relied on preexisting technology (such as labeling DNA with chemicals for optical analysis), while others employed new approaches, such as electronics.[47] Still others used an "atomic detection device" whereby a specialized microscope was used to examine base pairs identified by heavy metal atoms.[48] One analyst described this part of the market as "still very fragmented, with multiple companies developing technologies."[49]

One observer noted that with fewer pre-sequencing activities, "costs are lower and sequencing biases or errors can be reduced. A second beneficial feature [. . .] is that many of these technologies are long read length,[f] which improves the throughput [. . .]."[50] The ability to sequence long read lengths was an important factor in developing new technologies. While short read lengths were helpful for detailed examination of small sections of a genome, it was hard to know where each section fit within the order of the larger genome.[51] As one expert explained, "The longer read lengths have fundamentally more information [. . .] infinite coverage with short reads simply won't be enough for resolving really complex regions [of the genome], but just a few long reads in the right spot can solve them."[52]

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[e] In 1990, the U.S. National Institutes of Health and the U.S. Department of Energy, along with a number of partner entities from around the world, launched the Human Genome Project to sequence the 3 billion base pairs of the human genome. It was completed in April 2003. (Source: "Human Genome Project," *U.S. Department of Health & Human Services*, updated October 2010, report.nih.gov/NIHfactsheets/Pdfs/HumanGenomeProject%28NHGRI%29.pdf, accessed August 2013.)

[f] Read length referred to the number of base pairs in the segment of DNA that the machine was sequencing.

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One type of emerging technology was single-molecule sequencers, which did not require that DNA be divided into small segments and copied.[53] Such machines historically had high error rates, with one observer noting that "too often it sees the wrong nucleotide [. . .] in a DNA strand."[54] Moreover, individual machines that relied on the same underlying technology, could perform very differently. In one test, researchers reported error rates of under 2% for the Ion Torrent PGM, less than 1% for a range of Illumina's NGS machines, and nearly 13% for Pacific Biosciences' RS sequencer.[55] While the error rate for the RS was high, researchers believed it was useful for its long read lengths and could eventually become an important technology.[56]

Furthermore, the reliability of single-molecule sequencers was improving.[57] For example, a software product designed in 2012 helped address some problems.[58] One team of researchers noted that while one single-molecule sequencer had error rates of over 10% on single reads, errors were not frequently repeated between runs, so when the DNA was sequenced multiple times the device could produce a "highly accurate statistically averaged consensus perspective" genome.[59] These machines were expected to become commercially available in 2013 or 2014.[60] Companies in this market segment included Pacific Biosciences and Oxford Nanopore.

In addition to these new technologies, a number of Sanger-based machines were still in use, though new purchases were declining (see **Table A**).[61] (See **Exhibit 6** for company share of the total sequencer market, including machines from all generations in use, as of late 2012.)

Table A Revenue of Sequencers by Technology Generation, (US$ millions)

Platform	2010	2011	2016 (forecast)
Sanger	$213.1	$193.7	$91.9
NGS	396.0	564.7	909.8
Emerging	-	-	77.5
Total	**$609.1**	**$758.4**	**$1,079.2**

Source: Stephen Weber, "Top Ten Companies in DNA Sequencing," *BCC Research*, October 2012, p. 20, via BCC Research, a publisher of technology market research reports based in Wellesley, Massachusetts, accessed August 2013.

Choosing a Sequencer

Different machines helped meet customers' specific sequencing purposes.[62] Devices varied in how they sequenced DNA, the amount of information processed at one time, perceived quality, and other measures.[63] A sequencer's price and size were other important considerations. Also important was what type of—and how much—data a customer wanted. When researching and sequencing a new genome for example, highly-skilled scientists sequenced small DNA segments of the same genome multiple times and then examined them in order to create a complete picture of the entity's genome (this was called *de novo* sequencing).[64] A user's needs were different once a reference genome was built through *de novo* sequencing; subsequent sequencing of that genome (resequencing) focused on looking for specific attributes within that genome, such as comparing multiple different DNA samples to find and give meaning to genetic anomalies.[65]

One expert provided illustrations of those two methods in practice: "Sequencing applications in [agricultural biotechnology] include *de novo* sequencing of plant and animal species that are important for food or industrial use to create reference genomes. After these reference sequences are created, it is important to understand the correlation between genetic variation and important traits such as drought or pest resistance in crops. This step involves resequencing."[66] Even though *de novo*

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sequencing had to be done relatively few times, "Much of the sequencing done today is still *de novo*, as the number of species yet to be sequenced is large," the same expert noted.[67]

Customers, therefore, chose sequencers that matched their needs. For example, benchtop sequencers were useful for examination of a few genes to find specific, known mutations, and were thus important to clinical customers such as those performing diagnostic tests.[68] Higher-throughput machines were better suited to examining an entire genome.[69] "For medical and diagnostic resequencing applications, the overall cost of a test is more important than the cost per base sequenced. Accuracy and run time are also very important for these applications. *De novo* sequencing places a premium on the cost per nucleotide sequenced since the number of bases sequenced is usually large. Accuracy is important, but sequencing coverage makes up for lower accuracy in *de novo* sequencing," one expert elaborated.[70]

Because of this, device manufacturers were not automatically competitors.[71] For example, one category of Illumina's and Life Technologies's instruments were machines of choice in clinical settings, while Roche's and Pacific Biosciences's technologies were popular for research.[72] Institutions that specialized in sequencing-as-a-service could own many different devices.[73] This made them more attractive to potential customers and helped account for each device's shortcomings.[74]

Nor were old technologies made irrelevant by newer generations. As one analyst noted, "Sanger sequencing will continue to fill a niche market not addressed by the new sequencing formats. For example, in diagnostics, Sanger sequencing is a reliable method to confirm NGS results. Also, Sanger sequencing is used for single-gene disorders to confirm a diagnosis or detect the familial variant. In research applications, Sanger sequencing will continue to be used in smaller sequencing projects that don't need whole genomes or whole exomes [the parts of DNA that made proteins]."[75] Some NGS technologies, such as Roche's GS FLX+ and the Ion PGM, were approved for research applications, but not yet for clinical diagnostics.[76]

A machine's efficacy for a given application was influenced by how DNA was prepared for and read by a sequencer. For example, one group of researchers found that the Ion Torrent platform was less adept at sequencing genomes made up of adenine and thymine (AT) base pairs, which could be an important factor for those working in specific fields, such as the study of malaria, as the researchers noted that over 80% of the malaria parasite's genome was composed of AT base pairs.[77] After testing a number of NGS platforms, the researchers concluded that, "The decision on whether to purchase one or the other [sequencer] will hinge on available resources, existing infrastructure and personal experience, available finances, and the type of applications being considered."[78]

Purchasing decisions were typically made with input from multiple stakeholders. Researchers and those that would use the sequencer(s) typically had a good deal of influence around which device(s) to buy, though a purchasing agent would likely manage the process, while larger institutions such as hospitals and diagnostic labs could have purchasing departments empowered to make this decision. "It's generally more nuanced than this though," Rastegar said. "The business development person, for example, might be the decision maker around a pilot or beta test with a new sequencing company, but then it's the scientists who perform experiments and make the recommendation of whether to adopt a new technology based on their experience."

Founding GenapSys

A native of Iran, Esfandyarpour grew up in a family passionate about learning. "My mother said that our goal should be to use our education to give back to our country, to do something good for

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the people. I've always held that as my life philosophy," he explained. GenapSys was the culmination of a lifelong mission for Esfandyarpour. He had experienced the limitations of modern medicine firsthand through the catastrophic medical misdiagnosis of a relative. He believed his technology could lead to more research into disease genetics and be used for diagnostic testing.

Esfandyarpour graduated from Iran's Sharif University of Technology in 2000. He discovered his entrepreneurial talents while still a student when he founded an Internet service provider. Its success proved to be a pivotal moment. "I knew that money would never be the main driver for me. I could always make money if I needed to and wouldn't die of hunger. This realization enabled me to take risks and go after meaningful opportunities," he explained.

After graduation, Esfandyarpour left to study electrical engineering at Stanford University. He initially focused on telecommunications, but with an interest to have a more direct impact on humanity, he explored several academic departments. Through a conversation with a friend working at the Stanford Genome Technology Center (SGTC), he learned about the optical method for DNA sequencing. It struck him that there might be a way to replace this complicated process with electronics to sequence DNA faster and cheaper.

Esfandyarpour approached the SGTC's director, Professor Ron Davis. "Davis knew how powerful this technology could be. He convinced me to work on it as my PhD project and that's how the journey began back in 2004." Between 2004 and 2009, Esfandyarpour worked on the technology that would become the GENIUS device, leveraging the extensive resources and grant funding the university and the STGC secured for his project, including tens of millions of dollars from the National Institutes of Health (NIH). Esfandyarpour successfully defended his PhD in 2009 and incorporated GenapSys in January 2010. He also assembled an advisory board of leading experts in the field (see **Appendix A**).

Between 2010 and 2013, GenapSys was awarded $4 million in NIH grants to develop its device further. The company raised some $3 million from angel investors during this time as well. In April 2013, GenapSys secured $10 million in venture funding with Decheng Capital as the lead investor. IPV Capital and Stanford University also invested in the Series A round (see **Appendix B** for information on GenapSys's investors). Management planned to raise $20 million to $40 million in capital in late 2013. GenapSys had a monthly burn rate of $700,000, and the new capital would carry the company through product release, commercialization, and until it became cash flow positive.

By 2013, the company was on its third generation of the device, which measured electrical changes through semiconductor-based nanosensors. DNA was loaded onto a microchip in a cartridge that was then inserted into the back of the machine (see **Exhibit 7**). GenapSys would offer three test cartridges with microchips of varying capacity: 1M (one gigabase pair (gbp)[g] of throughput), 20M (20 gbp of throughput), and 100M (100 gbp of throughput). The different cartridges would be released over time—the 1M first and the 100M last—but would have the same price points.

The sequencing process required no further human intervention after the cartridge was inserted. Both the processed sequencing results and the raw data (which the GENIUS device required less of to arrive at meaningful results than other technologies) were uploaded directly to a cloud-based storage

[g] A gigabase was defined as "One billion bases (nucleotides) as a unit of length of a nucleic acid." (Source: Wordnik "Gigabase," https://www.wordnik.com/words/gigabase, accessed December 2014.)

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system, thus eliminating the need for onsite servers to store data.[h] The cost to sequence an entire human genome on the GENIUS machine could be as low as $100 and completed within eight hours. The GENIUS device had an expected error rate of less than 0.1% based on internal test results.

GenapSys's system was much smaller than other devices, with roughly the same footprint as an iPad (see **Exhibit 8**). While size was not as important to large research centers, it was a key consideration for customers such as hospitals and labs where space was scarce.

Competitive Landscape

Illumina and Life Technologies (specifically its Ion Torrent platform) were GenapSys's top competitors (refer to **Exhibit 3** and see **Exhibit 9**). Illumina's portfolio of sequencers varied in price from $125,000 to $740,000. Illumina's HiSeq line of sequencers were large, high-output machines while its smaller MiSeq sequenced less DNA in a single run.[79] HiSeq sequencers were widely used by prominent research and sequencing centers such as the Broad Institute.[80]

Life Technologies's Ion Torrent platform included both the Ion Torrent Proton and the Ion Torrent PGM. The Ion Torrent Proton (priced at $150,000) launched in early 2012 and was capable of sequencing an entire human exome or genome.[81] The Ion Torrent PGM (priced at $50,000) was better suited for smaller projects, such as sequencing a single gene.[82] However, these two sequencers required additional equipment (such as data servers, nitrogen tanks, and a water purification system), which could bring the overall cost of the devices close to $225,000 and $105,000, respectively.

Life Technologies's Ion Torrent platform was the only other device similar to GenapSys's machine in that it also used semiconductor-based microchips to sequence DNA.[83] A number of emerging companies, such as Oxford Nanopore, Genia, and Nabsys, were working on electronic sequencing platforms as well, though they were all single-molecule sequencers. Oxford Nanopore was working on its MinION sequencer, a handheld $900 dollar disposable sequencer it had not yet commercially launched.[84] One analyst valued Oxford Nanopore at $1.5 billion in early 2013.[85]

While companies with promising technology could generate considerable buzz, some had not been able to translate this into commercial success. Helicos Biosciences, a Massachusetts-based company with an optical single-molecule sequencer, had raised tens of millions of dollars before launching its machine.[86] The company had only 14 customers by 2010, and an analyst noted that the device's price (at $800,000) inhibited its growth.[87] The company filed for Chapter 11 bankruptcy protection in late 2012.[88]

The Business Models

As GenapSys moved closer to launching its device, Esfandyarpour and Rastegar discussed the many possible ways to bring its sequencer to market. They narrowed their choices to three options.

Traditional Sales Strategy

The most straightforward option was to follow other sequencing companies and sell to major research institutions, pharmaceutical firms, and labs offering diagnostic tests, where there was

[h] Rastegar noted that this was a competitive advantage as "In other systems, huge data storage and processing requirements force users to destroy the raw data and only deal with the processed data. This means that the data they have becomes incompatible over time for data analysis as algorithms improve."

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already demonstrated demand. The device's low cost would attract entities unable to afford traditional, higher-priced sequencers as well. Sales and distribution channels were well established and GenapSys would run a small sales and support team of fewer than 10 people. The company would sell relatively few devices but earn high margins per machine given its low cost design.

Rastegar identified a market of a few thousand potential customers, representing demand for about 2,000 devices. Because of its low manufacturing costs, GenapSys could price its machine below those of the major incumbents. The machine would initially be priced at $40,000, though this would gradually decline over time. Additional revenues would come from selling test cartridges (at $500 to $1,000 each) and from servicing machines every other year at a cost of $10,000 (each time). The goal was to price the device and cartridges lower than competing products, but still high enough to generate substantial revenue in a short timeframe.

This model was risky as it directly challenged incumbents. "Our target customers would be those served by Illumina and Life Technologies and there is no way they wouldn't retaliate. We leave ourselves ripe for competition by going for just a slight reduction in price," Esfandyarpour explained. At $50,000, the Ion PGM was priced just above that of the GENIUS platform. Rastegar recalled one company that tried to undercut a major incumbent on price in the early 2010s. The incumbent simply dropped their price below that of the new company until it had been sufficiently marginalized, and then raised prices when the competitor was no longer a threat. It was also unclear if customers would switch to GenapSys's instrument if they already had other NGS machines in place. Incumbents might start introducing incentives to stop customers from moving over to GenapSys.

Esfandyarpour and Rastegar also considered the financial stability of this market. To pay for sequencers, most research centers relied on grants and government funding, which could be unreliable. For example, when Congress could not decide on a budget in 2013 and mandatory cuts came into effect, the NIH, a source of funding for researchers, saw its budget cut by $1.55 billion.[89] Furthermore, what type of sequencing device(s) an entity had in place could affect whether or not it received funding, with one analyst noting a preference for NGS technologies.[90]

GenapSys's key value proposition under this strategy was not just its price, but what management termed the GENIUS machine's "four advantages"—its low cost, small size, ease of use, and speed (and the increased productivity this enabled). Even leading research institutions were limited in their activities by lengthy sequencing times and had to schedule machines well in advance of their actual use. Large scale research projects could wait for months before enough machines became available. GenapSys could position its device as a tool just as accurate as the leading sequencers, while enabling customers to receive faster results, complete more projects, open up new partnership opportunities, and take on additional grant money.

Razor-and-Razor-Blade

The second option was to follow a "razor-and-razor-blade"[i] strategy: price the machine between $5,000 and $7,000 to bring in a high volume of customers, and charge $200 for test cartridges (on which GenapSys earned a roughly 50% profit margin). At this price, new markets would open for GenapSys, including small research labs and biotechnology (biotech) firms, hospitals, and universities. Diagnostic testing companies and labs would also be interested. National laboratory companies, such as Quest Diagnostics, which performed high volumes of tests for hospitals and

[i] This strategy referred to the way shaving razors and their replacement blades were priced. While the initial razor was sold at a low price, replacement blades were more expensive and their sales generated more revenue than the razors.

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clinics, would also be targeted. While both the device and test cartridge price were lower in this model than in the traditional one, management hoped that increased testing volume would drive revenues. This concept was endorsed by one expert: "The sale of a sequencing instrument generates a future stream of high-margin product revenue from sample preparation reagents, sequencing consumables and bioinformatics. This revenue stream can last for several years until the next instrument-purchasing cycle occurs for a customer. Therefore, having an installed base of sequencers is a key success factor for instrument companies."[91]

Management believed there was significant pent up demand for affordable sequencers. Just 3,000 of the 50,000 molecular research labs operating globally in 2011 owned a sequencer or could afford to purchase one in the near future, analysts had observed.[92] On the clinical side, one market research firm believed that just over 2,000 entities around the world (primarily hospitals) would ultimately have sequencers, and that 25% of this number would have them by 2015.[93] However, just because companies could afford to purchase sequencers did not mean they would. One analyst believed that falling device and sequencing prices would lead to a "democratization of sequencing," but also went on to note that "[the] need for outsourced sequencing services done by third-party laboratories will continue to grow due to the high instrument and infrastructure investment required for NGS systems, which can be an issue for smaller companies and [labs]."[94] Small pharmaceutical and biotechnology companies were particularly likely to outsource.[95] (See **Exhibit 10** for revenue for sequencing services by industry.)

GenapSys would also target players in industries relatively new to sequencing, such as biofuels companies and agribusiness firms, to help them better understand complex plant genetics. Rice was made up of between 46,000 and 55,600 genes, for example, while bread wheat DNA had some 17 billion base pairs.[96] Some firms had already engaged in sequencing. In 2002, Syngenta and its partners sequenced the genomes of two subspecies of rice, and in 2009, Bayer Crop Science of Bayer AG announced that, in collaboration with its partners, it had sequenced the genome of the rapeseed plant.[97] The agribusiness company Monsanto had taken these partnerships a step further and invested directly in the device-manufacturer Pacific Biosciences.[98]

Management believed that by focusing on customers new to sequencing, as opposed to targeting established users served by established players, larger firms such as Illumina and Life Technologies would not see GenapSys as a direct threat. "We'd simply be expanding the market to include entities that otherwise could not afford sequencers instead of eating into their market share," Rastegar explained. Instead, competition would primarily be with other emerging technologies.

This model opened the possibility for GenapSys to focus on underserved markets and form partnerships with biotech or diagnostics companies around researching and better understanding a certain disease or disease category. "Our technology could be an enabling tool for solutions for a specific patient group," Esfandyarpour elaborated. However, similar partnerships had generated controversy, particularly when these agreements were seen to limit others' research.[99] If GenapSys or its partner exercised too much control over the intellectual property generated through such a partnership, there was a risk this could generate controversy. Such was the case when the biopharmaceutical firm Geron Corporation exclusively licensed the ability to develop products from six types of stem cells from the University of Wisconsin's Wisconsin Alumni Research Fund.[100] Some believed these patents made it challenging for outside parties to use and study stem cells.[101]

Because this model relied on cartridge sales, the pressure to quickly scale and reach a critical mass of customers loomed large. "With our capital expenditure committed to building manufacturing capabilities to meet the anticipated demand, we would be in danger of not breaking even if we don't

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sell enough of them," Rastegar explained. GenapSys could potentially have a sales team of 20 to 30 people to reach this broader range of customers. The company could also use existing sales channels, partners, or sign distribution agreements with strategic enterprises. Management wanted to be careful structuring agreements that might limit growth, and decided not to license its technology.

Data Analytics

The third model focused not on selling machines, but on the data these machines generated. "The same market forces that are making sequencing more accessible will commoditize the market as firms compete on price," Rastegar explained. If GenapSys could collect large sets of aggregated genetic data for multiple diseases from many individuals, this information would be useful for research, commercial R&D, and designing new diagnostic tests.

Other entities recognized the value of large datasets as well. Boston Children's Hospital and the nearby Brigham and Women's Hospital announced that in 2014, the two hospitals would begin a five-year project to sequence the DNA of 480 infants and their parents to see how this information affected the childrens' medical care.[102] In 2012, the British government announced it was collecting the genomes of 100,000 patients with cancers and rare diseases through the country's National Health Service (NHS).[103] "By unlocking the power of DNA data, the NHS [. . .] will lead the global race for better tests, better drugs and above all, better care," explained the country's prime minister.[104] "The entity that puts this together and makes the correlations between data and disease will be the one that prevails. This is a classic network effect business," Esfandyarpour believed.

In this scenario, customers could buy a GENIUS sequencer for $3,000 or lease one for $2,000 per year, and purchase test cartridges for $50 to $100. Customers would give GenapSys access to a copy of the genomic data generated through the machines for analytical comparison purposes. This was the key source of revenue under this model as the sequenced genomes would be used in a database of genetic data. "We could become the Google of genomics," Esfandyarpour said.

Tests built from this data would be hosted on GenapSys's website in a format similar to Apple's App Store, and customers could order tests they would perform by running DNA through their own GenapSys machine. Company management believed the clinical diagnostic testing market would take off as sequencers became widely adopted. An analyst believed the market would grow from some $732 million in 2012 to $7.6 billion by 2018 on NGS machines alone.[105] "In the long run, this model is the way of the future. Everyone is trying to get here and our device could be the enabling technology," Esfandyarpour said.

Potential customers ranged from major biotech and pharmaceutical companies (to understand what part of the population might be receptive to a drug, for example), to government agencies, diagnostics companies, researchers, and physicians. Esfandyarpour ultimately wanted to expand the market to individual consumers to use the machine at home, perhaps as a way to monitor a health condition similar to how diabetics monitored their blood sugar levels. He also saw potential for the device outside of pure healthcare applications. "It will be important for environmental purposes and for purely educational uses. If students have access to this device they can sequence everything around them and help digitize the world," Esfandyarpour said.

GenapSys would primarily derive its revenues from test royalties (individual consumers would be charged a low fee per test), and from licensing or charging a subscription fee for access to its database. GenapSys could also form partnerships similar to those in the razor-and-razor-blades model. The genetic database and the online store would take time to develop, and these functions were expected to come online in 2015 or 2016.

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While this ambitious model had the greatest potential reward, it carried high risk. This strategy required significant R&D investments and created large overhead costs. Management wondered if the time and financial requirements would be too much for traditional investors. "We'd need investors with deep pockets who want to 'go for gold', and who recognize the massive opportunity that lies in owning such large amounts of data. Investors would have to be patient and comfortable with risk," Rastegar explained.

A large portion of the company's R&D would go towards software and bioinformatics (the platform that analyzed sequenced DNA and interpreted results) to support its genomic database and online store. "We're a technology development company, and downstream analytics isn't our core focus today," Esfandyarpour noted. The team had internally discussed partnering with a company experienced in handling large amounts of data such as Google or Amazon to help it scale.

Getting a genetic database up and running had a first mover advantage and if it were to go down this path, GenapSys wanted it in place before another major entity decided to enter the field. To get there, GenapSys needed a critical volume of customers to have sufficient genetic data in its database. The company planned to have a small sales force, believing that customer interest would be so great it could largely respond to incoming demand. To achieve this, management foresaw a need to use other marketing tools such as social media, and to have an e-commerce platform for customers to buy directly from GenapSys without the need for a direct sales force.

This model raised questions about patient privacy,[j] and data ownership. Despite the risks and uncertainties, Esfandyarpour and Rastegar believed this model had tremendous potential.

What Should GenapSys do?

As Esfandyarpour and Rastegar considered which model to present to GenapSys's investors, they recognized there was no obvious choice. Each model influenced how the company would develop in terms of its customer base (see **Exhibit 11**), sales and marketing, application of its technology, finances, future capital needs, and ownership. The two also recognized the need to balance the company's core value (to fundamentally change the way healthcare was delivered to patients) with investors' expectations for a strong financial return. Which model helped GenapSys reach the greatest number of customers and gave it the greatest likelihood for a secure financial future?

Esfandyarpour and Rastegar debated whether to pursue a hybrid strategy with elements from different models, or start with one model with the long-term goal of pursuing another strategy. The team was conscious that others were working towards building genetic libraries and other functions GenapSys wanted under its data analytics model. Did it risk giving up ground if it delayed acting? Esfandyarpour explained his views on each model: "The traditional model is a financially safe route. It's a proven strategy and there's a good chance we'll be acquired, but I don't see this as the way to change the future of medicine. With the razor-and-razor blade model, GenapSys can bring an affordable sequencing technology to new markets. However, it will ultimately be the company that controls the information that will succeed and change how healthcare is delivered."

[j] In the U.S., a patient's personal information was protected under the Health Insurance Portability and Accountability Act of 1996 (HIPAA). HIPAA required that patients consent to having their medical information shared in most situations outside of their normal care. (Source: "Your Health Information privacy Rights," U.S. Department of Health and Human Services, Office for Civil Rights, http://www.hhs.gov/ocr/privacy/hipaa/understanding/consumers/consumer_rights.pdf, accessed October 2013.)

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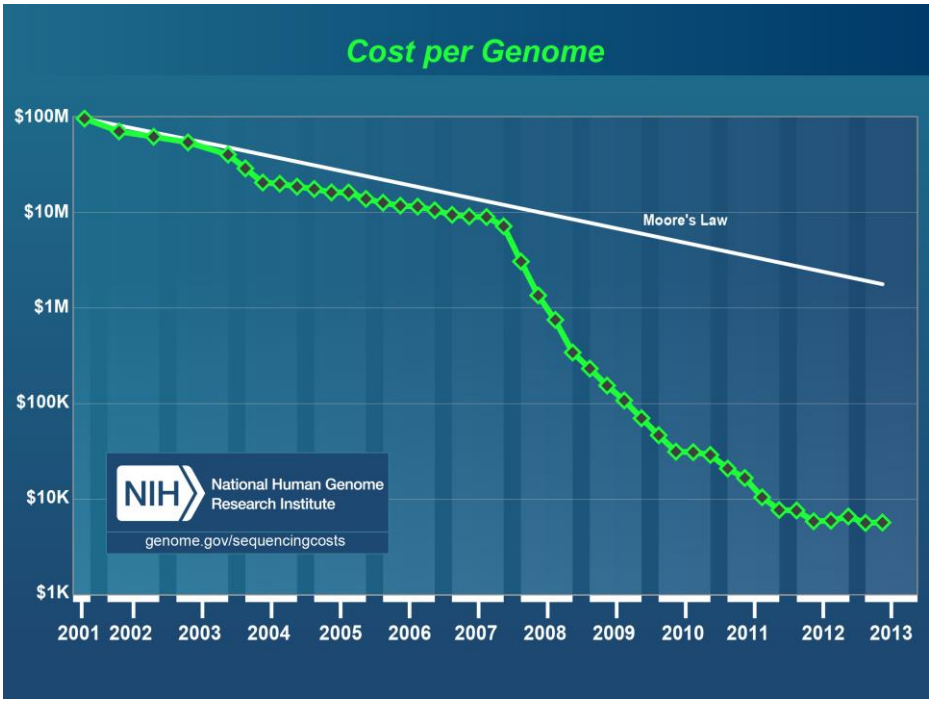
Exhibit 1 Diagnostic Testing Revenue by Category (in US$ millions; 2018 figures are forecasts)

	2011	2012	2013	2018
Infectious diseases	$5,564.4	$6,529.8	$7,664.8	$14,600.0
Oncology	$4,919.5	$5,837.7	$6,942.3	$13,000.0
Pre/post natal screening	$3,599.2	$4,030.6	$4,514.6	$7,410.0
Identity/paternity	$553.7	$604.7	$660.5	$930.0
Other	$304.4	$333.8	$366.0	$550.0
Total	**$14,941.2**	**$17,336.6**	**$20,148.2**	**$36,490.0**

Source: Jon Evans, "Molecular Diagnostics: Technologies and Global Markets," *BCC Research*, March 2013, p. 74, via BCC Research, a publisher of technology market research reports based in Wellesley, Massachusetts, accessed October 2013.

Exhibit 2 Cost to Sequence a Human Genome, 2001–2013



Source: "DNA Sequencing Costs," *National Human Genome Research Institute*, updated July 16, 2013, http://www.genome.gov/sequencingcosts/, accessed August 2013.

Note: Moore's law was "an axiom of microprocessor development usually holding that processing power doubles about every 18 months especially relative to cost or size." (Source: "Moore's law," *Merriam-Webster*, http://www.merriam-webster.com/dictionary/moore%27s%20law, accessed October 2013.)

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Exhibit 3a Data on Select NGS Systems, and One Sanger System

	454 GS FLX+ (Roche)	PacBio RS	Illumina GAIIx	Illumina HiSeq 2000	Sanger 3730 xl
Cost[a]	$500,000	$695,000	$256,000	$654,000	$95,000
Sequencing Yield Per Run	0.7 Gb	100 Mb	30 Gb	600 Gb	1.9 Kb to ~84 Kb
Sequencing Cost Per Gb	$7,000 (cost per run)	$2,000	$148	$41	$4 per 800 base pair reaction
Run Time	24 hours	2 hours	10 days	11 days	20 minutes to ~3 hours
Observed Raw Error Rate	0.1%	12.86%	0.76%	0.26%	0.001%
Read Length	700 base pairs	Average of 1,500 bases[b]	Up to 150 bases	Up to 150 bases	400 to ~900 base pairs

Source: Quail et. al. "A Tale of Three Next Generation Sequencing Platforms: Comparison of Ion Torrent, Pacific Biosciences, and Illumina MiSeq Sequencers," *BMC Genomics* 2012, 13:341, (published July 24, 2012), http://www.biomedcentral.com/content/pdf/1471-2164-13-341.pdf, p. 2, accessed May 2014.Data for the 454 GS FLX+ and the Sanger 3730 xl is from Lin Liu, Yinhu Li, Siliang Li, Ni Hu, Yimin He, Ray Pong, Danni Lin, Lihua Li, and Maggie Law, "Comparison of Next-Generation Sequencing Systems," *Journal of Biomedicine and Biotechnology* 2012, (accepted April 2, 2012), accessed via BioMed Research International, Hindawi Publishing Corporation, http://www.hindawi.com/journals/bmri/2012/251364/, May 2014.

[a] "All cost calculations are based on list price quotations obtained from the manufacturer and assume expected sequence yield stated," for the PacBio RS, the Illumina GAIIx, and the Illumina HiSeq 2000.

[b] "Mean mapped read length included adapter and reverse strand sequences. Subread lengths, i.e. the individual stretches of sequence originating from the sequenced fragment, are significantly shorter."

Exhibit 3b "Comparison of Benchtop Sequencers" 2012

	454 GS Junior (Roche)	Ion Torrent PGM (Life Technologies)	MiSeq (Illumina)
Cost	$108,000	$80,490	$125,000
Run Time	4 hours	3 hours	27 hours
Cost Per Run	$1,100	$425 (316 Chip)	$750
Throughput Per Run	71 to 72 mega base pairs	260 to 304 mega base pairs	1,653 mega base pairs
Average Read Length	522 base pairs	123 base pairs	2 x 150 base pairs
Reads Aligned	99%	90%	99%

Source: Dan Koboldt, "Comparison of Benchtop Sequencers," *MassGenomics*, April 27, 2012, http://massgenomics.org/2012/04/comparison-of-benchtop-sequencers.html, accessed May 2014.

Note: A megabase pair was a "Unit of length for DNA fragments equal to 1 million nucleotides [. . .]." (Source: Genetics Home Reference, "Megabase," U.S. National Library of Medicine, National Institutes of Health, December 16, 2014, http://ghr.nlm.nih.gov/glossary=megabase, accessed December 2014.)

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Exhibit 4 Companies' Dollar Share of the NGS Market in 2011 (in US$ millions)

Company	Revenues	Percentage (rounded)
Illumina	$369.9	65.6%
Life Technologies	$105.8	18.7%
Roche	$67.6	12.0%
Other	$21.4	3.8%
Total	$564.7	100.0%

Source: Stephen Weber, "Top Ten Companies in DNA Sequencing," *BCC Research*, October 2012, p. 10, via BCC Research, a publisher of technology market research reports based in Wellesley, Massachusetts, accessed August 2013.

Exhibit 5 Comparison of "Emerging" Sequencing Companies' Offerings, 2013

Company	Price of Machine	Read Length (in base pairs)	Throughput (in gigabase pairs)	Price per Run	Speed per Run[a]
GenapSys GENIUS[b]	$3,000 to $40,000	500	1-100	$100	<8 hours
Oxford Nanopore (GridIon)	$30,000	50,000	4	$5,000	6 hours
Oxford Nanopore (MinIon)	$900	50,000	1	$900	6 hours

Source: Company documents.

Note: A gigabase was defined as "One billion bases (nucleotides) as a unit of length of a nucleic acid." (Source: Wordnik "Gigabase," https://www.wordnik.com/words/gigabase, accessed December 2014.)

[a] This category included all the time involved from preparing a DNA sample through to receiving the "answer" from that sample, and not just the sequencing stage.

[b] The price per machine and price per run for GenapSys's device reflected potential prices. This device price included a 60% built in margin. These figures were subject to change depending on which business model management chose to pursue. The range in the throughput category was due to the three different microchips offered by GenapSys. The microchips had varying numbers of sensors, which gave customers the flexibility to process anywhere from 1 gigabase pair to 100 gigabase pairs of DNA.

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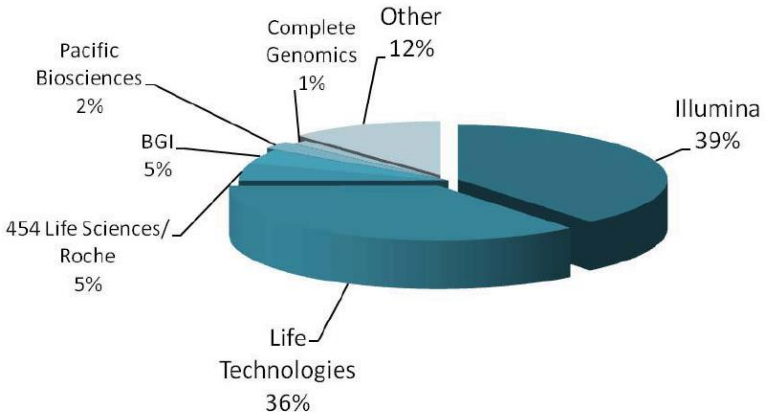
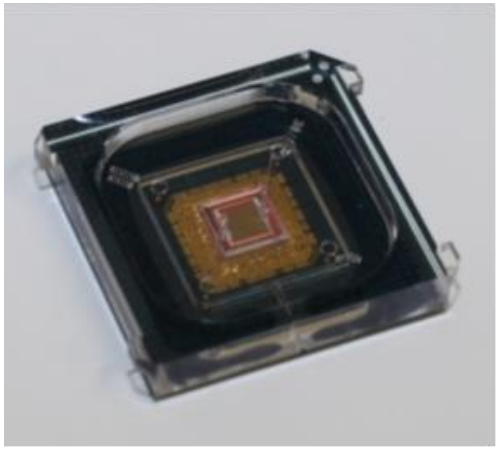
Exhibit 6 DNA Sequencer Market Share by Company, September 2012 (by number of instruments installed)



Source: Justin Saeks, "DNA Sequencing and Equipment Markets, 3rd Edition," *Kalorama Information*, September 2012, p. 97.

Exhibit 7 The GENIUS Platform Microchip



Source: Company documents.

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Exhibit 8 Size Comparison of Life Technologies' Ion Torrent PGM (the sequencer on the left; including a server and nitrogen tank) and the GENIUS machine (the sequencer on the right)

Source: Company documents.

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Exhibit 9 Select Financial Data from GenapSys's Main Competitors (US$ millions)

(52 weeks; December, 30 2012)	Illumina
Total revenues	**$1,148.5**
Selling, general, & administrative	$230.6
Research and development	$231.0
Total operating expense	**$947.8**
Net income	**$151.3**

2012 Annual Report	Life Technologies
Total revenues	**$3,798.5**
Selling, general, & administrative	$1,054.6
Research and development	$341.9
Gross profit	**$2,134.5**
Net income	**$430.5**

Source: Life Technologies Corp., 2012 Annual Report, March 15, 2013, pp.31, 35, 36, http://www.lifetechnologies.com/content/dam/LifeTech/Documents/PDFs/about-us/annual-review/2012Annual-Report.pdf, accessed December 2014; and "Illumina Inc.," Annual Income Statement, December 30, 2012, OneSource Information Services, Inc., accessed December 2014.

Exhibit 10 Revenue from Sequencing Services (by industry, in US$ millions)

Industry	2011	2016	CAGR (%)
Clinical diagnostics	$738.3	$2,719.5	29.8
R&D, pharmaceutical, biotech	$207.1	$593.6	23.4
Applied	$38.0	$159.7	33.3
Consumer	$4.2	$55.8	67.8
Total	**$987.6**	**$3,528.6**	**29.0**

Source: Stephen Weber, "Top Ten Companies in DNA Sequencing," *BCC Research*, October 2012, p. 21, via BCC Research, a publisher of technology market research reports based in Wellesley, Massachusetts, accessed August 2013.

Note: The "Applied" category included such uses as forensics and agricultural genomics.

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Exhibit 11 Potential Markets for the GENIUS Device

Customer	What it used sequencers for	Approximate prices it would pay for a sequencer	Average number of runs performed on each device	Number of devices each customer could purchase
Academic institution	Research, education, and grant projects.	<$50,000	4 runs per week	1–5 devices
Agribusiness/ biofuel	Crop improvements (e.g., more insect-resistant or able to grow with less water).	$50,000–$100,000	1 run per day	100s of devices (for the largest firms)
Government	Government-funded genome projects; forensics; national security.	<$50,000 for genome projects; $10,000 to $15,000 for forensics and security purposes	2–3 runs per day, depending on the project	100s-1,000s of devices, depending on the project
Hospital	Diagnostic tests it chose to perform in-house.	<$25,000	4 runs per week	1–5 devices
Individual customer	Monitoring one's own personal health.	~$1,000	1 test per month (for an individual monitoring a disease)	1 device
Diagnostic lab, large (e.g., Quest Diagnostics)	Diagnostic tests (for such things as cancer, diabetes, HIV, pre-natal tests, etc.) on samples sent by hospitals and medical clinics.	$50,000–$100,000	2 runs per day (depending on overnight shifts)	100s–1,000s of devices (for use in its facilities nationwide)
Diagnostic lab, small	Diagnostic tests (for such things as cancer, diabetes, HIV, pre-natal tests, etc.) on samples sent by smaller partnerships with healthcare providers.	$15,000–$20,000	1 run per day	5–10 devices
Pharmaceutical firm	Clinical trials (for patient selection and monitoring), drug development, and companion diagnostics.	$50,000–$100,000	3 runs per week	Varied depending on usage. It could be in the 100s if used for drug development, or in the 1,000s if used in clinical trials
Research lab, large	Research projects to advance scientific discoveries. These labs did not perform diagnostic tests.	<$700,000	1 run per day	20–30 devices
Research lab, small	Same as Research lab, large.	<$100,000	1 run per day	1–10 devices

Source: Company documents.

Notes: The numbers given in the fourth column are per machine. Additionally, one run was not necessarily equal to one sample as multiple DNA samples could be processed in a single run (with samples barcoded for identification).

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Appendix A: The Scientific Advisory Board

George M. Church George M. Church is a Professor of Genetics at Harvard Medical School and Director of the Center for Computational Genetics. With degrees from Duke University in Chemistry and Zoology, he co-authored research on 3D-software & RNA structure with Sung-Hou Kim. His PhD from Harvard in Biochemistry & Molecular Biology with Wally Gilbert included the first direct genomic sequencing method in 1984; initiating the Human Genome Project then as a Research Scientist at newly-formed Biogen Inc. and a Monsanto Life Sciences Research Fellow at UCSF. He invented the broadly-applied concepts of molecular multiplexing and tags, homologous recombination methods, and array DNA synthesizers. Technology transfer of automated sequencing & software to Genome Therapeutics Corp. resulted in the first commercial genome sequence (the human pathogen, H. pylori, 1994). In 2005 he initiated the Personal Genome Project. [. . .] He has served in advisory roles for 12 journals, 5 granting agencies and 22 biotech companies. Current research focuses on integrating biosystems-modeling with personal genomics & synthetic biology.

Ronald W. Davis Ronald W. Davis is a Professor of Biochemistry and Genetics and Director of the Stanford Genome Technology Center. He is a world leader in the development of biotechnology, especially the development and application of recombinant DNA and genomic methodologies to biological systems. Early in his career, Davis developed the quantitative analysis of DNA by electron microscopy as well as the R-loop technique for mapping coding RNAs by electron microscopy. Davis was instrumental in developing phage lambda-based cloning vectors and showed how they could be used for large capacity cloning with both bacterial and eukaryotic DNA. He is a pioneer in the area of genomics and high throughput biochemical techniques. Pioneering many of the early techniques developed using recombinant DNA and genetic linkage analysis. In addition to his work in academia, Professor Davis has been involved in the spin-off of multiple start-up companies in the area of genomics from his laboratory at Stanford. Davis is a member of the National Academy of Sciences. He has served on the National Institutes of Health (NIH) Genome Research Review Committee, and as chairman of the World Health Organization Strategic Research Steering Committee.

Luke P. Lee [Lee] is [the] Arnold and Barbara Silverman Distinguished Professor of Bioengineering at UC Berkeley, the Director of the Biomedical Institute of Global Healthcare Research & Technology (BIGHEART) and a Co-Director of the Berkeley Sensor & Actuator Center. He was Chair Professor in Systems Nanobiology at the Swiss Federal Institute of Technology (ETH, Zurich). He received his [BA] in Biophysics and [PhD] in Applied Science & Technology: Applied Physics (major) / Bioengineering (minor) from UC Berkeley. [. . .] Prof. Lee has authored and co-authored over 250 papers on bionanophotonics, microfluidics, single cell biology, quantitative biomedicine, molecular diagnostics, optofluidics, BioMEMS, biosensors, SQUIDs and SERS.

Ali Mani Ali Mani is an Assistant Professor at the Flow Physics and Computation division of Mechanical Engineering Department of Stanford University. His current research is in the area of fluid mechanics. His thesis work involved investigation of optical effects by compressible turbulence. Through his [PhD], he investigated fluidic transport regimes in hybrid microchannels/nanochannels networks. His publications span across different disciplines in fluid mechanics including aero-optics, microfluidics, and aero-acoustics. He received his MS and PhD degree in mechanical engineering from Stanford University (2004, 2009 respectively) and his BS from Sharif University of Technology in Iran (2002).

Steve Quake Steve Quake is a Professor of Bioengineering at Stanford University, Co-Chair at Department of Bioengineering and Investigator of Howard Hughes Medical Institute. He studied

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physics (BS, 1991) and mathematics (MS, 1991) at Stanford University before earning his doctorate in physics from Oxford University (1994) as a Marshall scholar. He then spent two years as a post-doc in Nobel Laureate Steven Chu's group at Stanford University developing techniques to manipulate single DNA molecules with optical tweezers. [. . .] Dr. Quake's interests lie at the nexus of physics, biology and biotechnology. Throughout his career, he has been active in the field of single molecule biophysics; he has focused on precision measurements on single molecules, and in 2003 his group demonstrated the first proof-of-principle that sequence information could be obtained from single molecules of DNA. [. . .] His contributions to the development of new biotechnology at the interface between physics and biology have been recognized by recent awards from the MIT Technology Review Magazine, Forbes and Popular Science.

Juan G. Santiago Juan G. Santiago is a Professor of Mechanical Engineering at Stanford University. He specializes in microscale transport phenomena and electrokinetics. He received his MS and PhD in Mechanical Engineering from UIUC. [. . .] His research includes the development of microsystems for on-chip electrophoresis, drug delivery, sample concentration methods, and miniature fuel cells. Applications of this work include genetic analysis, drug discovery, chemical weapon detection, and power generation. [. . .] He is on the editorial board of the Journal of Microfluidics and Nanofluidics, an Associated Editor of the journal Lab on a Chip, co-founder of Cooligy Inc., co-inventor of micron-resolution particle image velocimetry (Micro-PIV), and director of the Stanford Microfluidics Laboratory. He has also authored and co-authored 80 archival publications, authored and co-authored 170 conference papers, and been awarded 25 patents.

Michael Snyder Michael Snyder is the Stanford Ascherman Professor and Chair of Genetics and the Director of the Center of Genomics and Personalized Medicine. Dr. Snyder received his [PhD] training at the California Institute of Technology and carried out postdoctoral training at Stanford University. [. . .] His laboratory study was the first to perform a large-scale functional genomics project in any organism, and currently carries out a variety of projects in the areas of genomics and proteomics both in yeast and humans. [. . .] He sits on many international advisory boards and was a Co-founder of Protometrix, Inc., a protein microarray company that was purchased by Invitrogen in 2004, and a new company, Affomix, Inc.

Eric Topol Dr. Topol, a world-renowned geneticist, cardiologist, and researcher, is the Director of Scripps Translational Science Institute in La Jolla, California. He also serves as the Chief Academic Officer for Scripps Health, a Professor of Translational Genomics at The Scripps Research Institute, and was recently named The Gary and Mary West Chair of Innovative Medicine. Prior to joining Scripps, Dr. Topol spent much of his career at the Cleveland Clinic, where he served as chairman of cardiovascular medicine. [. . .]

The Scripps Genomic Medicine program and Translational Science Institute are bringing physicians and research scientists together to discover and define the genes that may influence disease. The goal is to determine how one person's genome may make them more susceptible than another to a given disease—then to take what is learned into clinical research, drug discovery programs and individualized patient care trials.

Dr. Topol was named the most influential physician executive in healthcare by Modern Healthcare magazine in 2012.

Source: All descriptions are verbatim from "Scientific Advisory Board," GenapSys website, http://genapsys.com/about-us/scientific-advisory-board/, accessed May 2014.

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Appendix B: Information on the Company's Investors

Decheng Capital Based in Shanghai, China, Decheng Capital (Decheng) was a young private equity firm founded in 2011.[106] Decheng was co-sponsored by both Bay City Capital (based in the U.S.) and DT Capital (headquartered in China) as the Decheng Capital China Life Sciences Fund I.[107] The firm had raised a total of $125 million in capital.[108] Decheng had five portfolio companies, including investments in three Chinese companies (two focused on medical devices, and one on medical equipment and diagnostic products), and two U.S. companies including GenapSys.[109] Its other U.S. investment was in GeneWeave Biosciences, a molecular diagnostics company focused on infectious diseases.[110] The firm's stated emphasis was life sciences investments in China.[111] One of the fund's founders, Min Cui, had been a lead investor in Ion Torrent Systems, Inc. with Bay City Capital, and this was a key consideration in GenapSys's decision to have Decheng Capital as a lead investor.

Bay City Capital was founded in 1997 in San Francisco, California and had invested $1.6 billion in some 90 companies over its history.[112] Its portfolio companies ranged across the life sciences spectrum, from biomedical and biotechnology firms to those working on pharmaceuticals and preventive medicines.[113] It had 28 portfolio companies as of 2013.[114] The firm had raised some $850 million since 2003.[115] Bay City Capital made 35 investments between 2008 and 2012, nearly all of them in North American companies.[116] The firm's most recent fund, Bay City Capital Fund V, was created in 2007 with $500 million in capital.[117] Of 27 investments made between 2008 and 2013 in the fund (including multiple investments in some companies), 16 investments were still active while the firm had fully exited six investments and partially exited five.[118] Of the investments made in this fund, 17 were in pharmaceutical companies, six were in medical device companies, and four in medical technology companies.[119] The fund had an internal return rate (IRR) of 12.3% and a multiple of 1.32.[120] In January 2010, the firm had invested in Ion Torrent Systems Ltd., a DNA sequencing technology company later purchased by Life Technologies for $725 million.[121]

DT Capital Partners (DT Capital) was founded in 2000 and based out of Shanghai, China.[122] DT Capital invested in firms across a range of industries, including clean technology, manufacturing, healthcare, and retail, with a focus on Chinese companies.[123] Since 2003, the firm had raised just over $800 million.[124] The firm typically committed a maximum of $20 million to a single investment, and held on to its investments for between two and eight years.[125] For example, the company raised some $355 million in 2008 for its DT Capital China Growth Fund and invested in eight companies between June 2008 and November 2009.[126] All eight investments were still active as of 2013.[127] The fund had a net IRR of 4.4% and a multiple of 1.14.[128] As of 2013, it had 74 active portfolio companies and managed some $500 million in its funds.[129]

IPV Capital IPV Capital was a Shanghai, China-based private equity firm founded in 2000.[130] The firm invested in a range of industries, from clean technology companies to firms working on software products or medical technologies.[131] It typically invested early in a company's life, and committed between $3 million and $15 million.[132] It held on to its investments for three to five years on average, and preferred a seat on the board of its portfolio companies.[133]

IPV Capital had 23 portfolio companies as of 2013: 10 in the semiconductor industry; nine (including GenapySys) in the software and services industries; and four clean technology firms.[134] Since 2003, it had raised $226 million.[135] In 2011, it had launched its IPV Capital II fund with $145 million to invest in startups and early stage clean technology, manufacturing, medical technology, semiconductor, and technology firms.[136]

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